

Project Initialization and Planning Phase

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Project Proposal (Proposed Solution) report

The proposal aims to develop an intelligent **Human Development Index (HDI) Prediction System** using Machine Learning to accurately predict the Human Development Index score and classify countries into **Very High, High, Medium, or Low** human development categories. The system addresses the limitations of traditional manual analysis by providing fast, reliable, and data-driven predictions based on key socio-economic indicators. The proposed solution integrates a trained Machine Learning model with a **Flask web application**, enabling users to obtain instant HDI predictions through an interactive and user-friendly interface.

Project Overview	
Objective	To develop a Machine Learning-based Human Development Index (HDI) Prediction System that accurately predicts the HDI score using indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per capita, enabling informed policy-making and development planning.
Scope	The project focuses on collecting HDI-related datasets, performing data preprocessing and visualization, training a Linear Regression Machine Learning model, and deploying it through a Flask web application for real-time HDI prediction and user interaction.
Problem Statement	

Description	Traditional methods of analyzing Human Development Index data are often time-consuming and require manual interpretation of multiple socio-economic indicators. This limits quick assessment of a country's development level and delays informed decision-making.
Impact	An accurate HDI prediction system helps governments, policymakers, researchers, and international organizations evaluate development levels, identify improvement areas, monitor progress, and make data-driven decisions for sustainable human development.
Proposed Solution	
Approach	Develop a supervised Machine Learning model using Linear Regression trained on historical Human Development Index data and integrate it into a Flask-based web application to generate real-time HDI predictions.
Key Features	• Machine Learning-based HDI prediction model. • Predicts HDI score and development category (Very High, High, Medium, Low). • Interactive web interface with user-friendly input forms. • Data visualization using Matplotlib and Seaborn. • Fast and accurate prediction results. • Model serialization using Pickle. • Easy deployment using the Flask framework

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor or T4 GPU (for model training)
Memory	RAM specifications	8 GB or above

Storage	Disk space for datasets, models, and application files	Minimum 10 GB free space (SSD recommended)
Software		
Frameworks	Python Web Framework	Flask
Libraries	Machine Learning & Data Processing Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Pickle
Development Environment	IDE	Jupyter Notebook, Spyder, Visual Studio Code, or PyCharm
Data		
Data	Source, size, format	Kaggle Human Development Index (HDI) Dataset, CSV format (Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, GNI Per Capita, HDI Score)